

FedSLoP: Memory-Efficient Federated Learning with Low-Rank Gradient Projection

ReasFlow Paper Copilot

Abstract. Federated learning enables a population of clients to collaboratively train machine learning models without exchanging their raw data, but standard algorithms such as FedAvg suffer from slow convergence and high communication and memory costs in heterogeneous, resource-constrained environments. We introduce FedSLoP, a federated optimization algorithm that combines momentum with low-rank random subspace projections of stochastic gradients, thereby reducing the dimension of communicated and stored updates while preserving optimization progress. On the theoretical side, we develop a detailed nonconvex convergence analysis under standard smoothness and bounded-variance assumptions. For the full-participation setting, we prove an explicit $\mathcal{O}(1/T)$ convergence rate in terms of the averaged squared gradient norm, with a stochastic variance floor that exhibits linear speedup in the number of clients and explicit dependence on the subspace rank and momentum parameters. We further derive an optimized harmonic step-size strategy that attains an essentially optimal trade-off between optimization, variance, and drift terms in the presence of low-rank projections. On the empirical side, we conduct extensive experiments on federated MNIST classification with heterogeneous data partitions, showing that FedSLoP substantially reduces communication volume and client-side memory while achieving competitive or better accuracy compared with FedAvg and representative sparse or low-rank baselines. Together, our results demonstrate that random subspace momentum methods such as FedSLoP provide a principled and effective approach to communication- and memory-efficient federated learning.

AMS subject classifications: 68T07, 90C25

Key words: Federated Learning, Communication Efficiency, Low-Rank Projection, Gradient Compression

1 Introduction

Federated learning (FL) enables collaborative model training across many edge clients without centralizing raw data, offering a principled way to respect privacy and bandwidth constraints. In cross-device deployments, the Federated Averaging (FedAvg) algorithm has emerged as the de facto workhorse, but its standard instantiations do not explicitly account for the tight memory and compute budgets of resource-constrained

clients. This paper introduces FedSLoP, a FedAvg-style method that incorporates random subspace optimization and momentum to reduce client-side memory usage while maintaining compatibility with existing FL workflows.

In the canonical cross-device FL setting, a central server repeatedly broadcasts a global model to a subset of clients, each client performs several local stochastic gradient steps on its private data, and the server averages the resulting model updates to form the next global iterate. This Federated Averaging (FedAvg) procedure is attractive due to its simplicity and communication efficiency, yet its behavior deteriorates in realistic environments with heterogeneous (non-IID) data and partial client participation, where local models can drift away from the global optimum and slow overall convergence. Existing theoretical analyses of FedAvg and its variants often require restrictive conditions—such as decaying step sizes or strong assumptions on the data distribution—to establish convergence guarantees, and typically focus on statistical efficiency rather than system-level considerations such as client memory usage. At the same time, edge devices participating in FL frequently operate under tight memory and compute constraints, making the storage of full-dimensional optimizer states and intermediate activations a significant bottleneck in practice.

A rich line of foundational FL algorithms builds upon FedAvg to mitigate client drift and improve convergence under data heterogeneity. FedProx augments local objectives with a proximal term that penalizes deviations from the current global model, stabilizing local updates at the cost of more complex local subproblems. SCAFFOLD introduces per-client and server-side control variates to correct client drift and can attain convergence behavior comparable to centralized SGD even under strong heterogeneity and partial participation [13]. Subsequent work develops unified convergence analyses of FedAvg and its momentum variants, proving linear speedup in the number of clients and relaxing bounded-gradient assumptions in both convex and nonconvex regimes [4, 15, 21, 25, 35]. These results provide guidance on step-size and local step choices but largely abstract away the system-level constraints of cross-device FL, including the memory footprint of optimizer states, auxiliary control variables, and per-iteration activations on clients.

In parallel, a substantial body of work has focused on communication-efficient distributed and federated optimization, employing gradient quantization, sparsification, low-rank compression, and error-feedback mechanisms to reduce the number of transmitted bits. Unbiased compressors such as QSGD [1] and biased schemes including Top- k sparsification or sign-based updates are often combined with error-feedback frameworks that provably recover convergence rates comparable to uncompressed SGD [3, 11, 14, 20, 23, 24]. Integrated methods like Qsparse-local-SGD blend local updates with aggressive compression to match the convergence of standard distributed SGD in smooth nonconvex settings [2, 5, 8, 9]. While highly effective at saving communication, these approaches typically keep the optimizer state full-dimensional and are thus not tailored to reduce the client-side memory footprint, especially for modern deep models where momentum buffers and auxiliary variables can dominate memory consumption.

A third strand of research studies subspace and low-rank optimization, primarily in

centralized training. Recent methods show that optimization trajectories for large models often lie in low-dimensional subspaces, and exploit this structure to reduce optimizer memory by updating only within carefully chosen subspaces while maintaining standard convergence guarantees [10]. Related work on efficient subspace algorithms for federated learning suggests that structured projections can be leveraged to cope with data heterogeneity while controlling complexity [41]. However, most existing subspace approaches are not explicitly designed for the client–server architecture and participation patterns of cross-device FL, and their implications for client memory usage, especially in the presence of momentum and multiple local steps, remain underexplored.

These strands of work highlight a persistent gap between algorithmic and system-level considerations in federated learning. Foundational FL methods refine the optimization dynamics of FedAvg to handle heterogeneity but often increase client-side state (e.g., control variates or proximal auxiliary variables), while communication-efficient methods significantly reduce transmitted bits without shrinking the dimensionality of optimizer states. Subspace optimization, in turn, offers a promising avenue for reducing memory but has not been systematically integrated into standard FedAvg-style workflows in realistic cross-device FL settings. Consequently, we lack FL algorithms that are simultaneously memory-efficient on clients, theoretically grounded under realistic stochastic conditions, and compatible with the simplicity and robustness that have made FedAvg the default choice in many applications.

FedSLoP is designed to bridge this gap by combining FedAvg-style training with random subspace optimization and momentum in a way that explicitly targets client-side memory efficiency. Unlike SCAFFOLD and related variance-reduction methods that maintain additional per-parameter control variates on clients [13], FedSLoP constrains both stochastic gradients and momentum to a random rank- r subspace that is shared within each communication round. This design reduces the effective dimensionality of the optimization trajectory and the associated optimizer states, while preserving the standard pattern of broadcasting a global model, running local updates, and averaging model deltas. At the same time, our convergence analysis in the nonconvex regime, under random Stiefel projectors, makes explicit how the convergence rate depends on the subspace ratio $\underline{\delta} = r/d$ and the number of clients, thereby quantifying the trade-offs induced by low-dimensional random projections in federated learning.

Concretely, in each communication round t the server samples a random orthonormal basis P_t from the Stiefel manifold $\text{St}(d, r)$, forms the rank- r projector $\Pi_t = P_t P_t^\top$, and broadcasts the pair (θ^t, P_t) to the participating clients. Each selected client i initializes its local model at $\theta_{i,0} = \theta^t$ and performs τ local steps of SGD with momentum, but both the stochastic gradients and momentum are restricted to the shared subspace: given a stochastic gradient $g_{i,s} = \nabla F_i(\theta_{i,s}; \xi_{i,s})$, the momentum is updated as $v_{i,s+1} = \mu v_{i,s} + \Pi_t g_{i,s}$ and the model is updated as $\theta_{i,s+1} = \theta_{i,s} - \eta v_{i,s+1}$. After completing the local loop, the client returns the full-model delta $\Delta_i^t = \theta_{i,\tau} - \theta^t$, and the server performs simple averaging $\theta^{t+1} = \theta^t + \frac{1}{|S_t|} \sum_{i \in S_t} \Delta_i^t$. By construction, when $r = d$ the projection Π_t becomes the identity, and FedSLoP reduces exactly to standard FedAvg with momentum, providing a natural

sanity check in both the theoretical analysis and empirical evaluation.

By constraining all local updates to a random low-dimensional subspace with ratio $\underline{\delta} = r/d$, FedSLoP reduces the effective dimensionality of both the optimization trajectory and the optimizer state on each client. Clients maintain only a single momentum vector updated through the projector, without any per-parameter auxiliary variables or control variates, resulting in an inherently memory-efficient approach that is particularly attractive for resource-constrained devices. At the same time, the algorithm retains the core FedAvg workflow—periodic broadcast, local computation, and model averaging—ensuring that it can be integrated into existing FL systems with minimal engineering changes while explicitly encoding low-dimensional structure in the training dynamics.

On the theoretical side, we establish a nonconvex convergence analysis of FedSLoP under full client participation and random Stiefel projectors. Under standard smoothness and bounded variance assumptions, and explicit step-size conditions that depend on the momentum parameter μ , the number of local steps τ , and the subspace ratio $\underline{\delta} = r/d$, we show that the averaged squared gradient norm admits an explicit convergence bound with an optimization term decaying as $\mathcal{O}(1/T)$ and a stochastic variance term decaying as $\mathcal{O}(T^{-1/2})$, with explicit dependence on the number of clients N . The resulting bounds exhibit linear speedup in N through a variance term of the form σ_L^2/N , and their explicit dependence on $\underline{\delta}$ quantifies how restricting updates to low-dimensional random subspaces impacts stochastic noise and convergence behavior. These results demonstrate that incorporating random subspace constraints into FedAvg-style training need not deteriorate the convergence order, while making the algorithmic trade-offs between memory savings and optimization fidelity transparent.

Empirically, we evaluate FedSLoP on a federated MNIST classification task, comparing it against standard FedAvg with momentum and several sparse or partial-update baselines on a Dirichlet non-IID split. Under a unified training budget of 100 communication rounds and a shared learning rate, FedAvg with momentum achieves the highest final test accuracy, while FedSLoP attains a very close final accuracy and clearly outperforms sparse or partial-update methods such as FedMef, NeuLite, and FederatedSelect. These observations support the view that FedSLoP is a practical, memory-conscious variant of FedAvg that preserves competitive predictive performance while offering a favorable trade-off between performance and structural efficiency in realistic federated settings.

In summary, this work contributes a memory-efficient, theoretically grounded, and empirically validated variant of FedAvg that operates in random low-dimensional subspaces while preserving the standard cross-device FL workflow. Our design explicitly targets client-side memory constraints without introducing complex auxiliary states, and our analysis and experiments jointly characterize the trade-offs between subspace dimension, convergence, and communication in federated optimization.

- **FedAvg-style random subspace algorithm with momentum.** We introduce Fed-

SLoP, a variant of FedAvg that constrains local optimization to a random rank- r subspace shared within each round via projectors sampled from the Stiefel manifold. The method keeps the standard pattern of broadcasting the global model, running local SGD with momentum, and averaging model deltas, but applies the projector to both gradients and momentum on each client. This design eliminates the need for additional per-parameter states such as control variates, thereby reducing the dimensionality of optimizer states while preserving the simplicity and robustness of FedAvg-style training.

- **Nonconvex convergence analysis under random Stiefel projectors.** We provide a detailed nonconvex convergence analysis of FedSLoP in the full-participation setting, assuming smooth objectives and bounded stochastic gradient variance. The resulting bound on the averaged squared gradient norm has an optimization term of order $\mathcal{O}(1/T)$ and a variance-dominated term of order $\mathcal{O}(T^{-1/2})$, with explicit constants that depend on the momentum parameter, the number of local steps, the subspace ratio $\underline{\delta} = r/d$, and the number of clients. In particular, the bounds exhibit linear speedup in the number of clients through a variance term proportional to σ_L^2/N , and show that constraining updates to low-dimensional random subspaces does not change the convergence order.
- **Empirical validation of memory- and communication-friendly design.** Through experiments on federated MNIST classification under a Dirichlet non-IID split, we compare FedSLoP to standard FedAvg with momentum and several sparse or partial-update baselines across a unified training budget. FedAvg with momentum achieves the highest final test accuracy, while FedSLoP reaches a very close final accuracy and maintains a consistent advantage over FedMef, NeuLite, and FederatedSelect, indicating that its memory-efficient, low-rank design offers a favorable trade-off between performance and resource usage in federated optimization.

2 Related Work

Federated learning (FL) has emerged as a pivotal paradigm for training models on decentralized data. The foundational Federated Averaging (FedAvg) algorithm established the standard workflow of periodic local updates and server aggregation. However, its performance under realistic data heterogeneity (non-IID data) poses significant challenges, including client drift and slow convergence [15]. To address these issues, algorithmic extensions like FedProx and SCAFFOLD were introduced, with the latter using control variates to correct for client drift and achieve convergence rates matching centralized SGD under extreme heterogeneity [13]. Theoretical analyses have been crucial, proving that linear speedup in the number of clients is achievable for FedAvg even under simultaneous non-IID data and partial participation [21, 35]. Further analysis in overparameterized neural networks suggests that the impact of heterogeneity diminishes with increased

model width [12], while a non-parametric view quantifies the benefits of federation under model and covariate shift [25]. Systems challenges like non-uniform client participation have also been addressed, with methods proposed to correct participation bias [33] and handle computational heterogeneity by allowing lightweight local updates [39]. For composite objectives with non-smooth regularizers, primal–dual frameworks like Federated Dual Averaging [37] and algorithms like FedCanon [42] and FedDR [29] have been developed, though their guarantees often involve residual errors or restrictive assumptions [40]. Personalization is a key direction to handle heterogeneity, with meta-learning approaches like Per-FedAvg [22] and distillation-based frameworks [30,32]. Other strategies include feature-alignment models [36] and decentralized personalized methods [19]. However, these methods often incur communication and memory overheads for maintaining personalized components, and their robustness under severe system constraints requires further investigation.

Reducing communication overhead is essential for scalable FL. Decentralized optimization protocols offer an alternative to centralized servers, enabling peer-to-peer communication. While basic methods like D-PSGD suffer from non-vanishing error under data heterogeneity, variance-reduced extensions like D^2 achieve rates matching centralized SGD [27]. A key insight is that removing data heterogeneity influence enhances the dependence on network topology [38], and gradient tracking can provide network-agnostic performance under certain conditions [34]. Gradient compression techniques are fundamental, and can be categorized into unbiased and biased (contractive) compressors [11]. Algorithm-agnostic lower bounds establish fundamental performance limits for distributed optimization with compression [8]. To counteract compression errors, error-feedback mechanisms like EF21 are critical, preserving convergence even with biased compressors [14,24]. Momentum integration further helps mitigate variance from heterogeneity [4]. For practical FL integration, frameworks like Qsparse-local-SGD unify local SGD with sparsification and quantization [2], while double-pass compression schemes like DoubleSqueeze apply error compensation to both upload and download directions [28]. Asynchronous decentralized protocols can also alleviate synchronization bottlenecks [16]. Adaptive compression strategies and near-optimal algorithms like ADIANA and NEOLITHIC have been developed to approach theoretical lower bounds [8,9]. However, these advanced methods often assume strong convexity or rely on unknown problem constants, limiting their practical deployment in heterogeneous, non-convex FL settings. Other notable compression schemes include sparsification with memory [23], unified biased compression frameworks [3], quantization methods like QSGD [1], low-rank compression like PowerSGD [31], and accelerated error-compensated methods [20]. Primal–dual methods also unify local training with unbiased compression [5].

Memory constraints on client devices necessitate efficient management of optimizer states and activations. Memory-efficient optimizer design often exploits low-dimensional structure. Methods like GaLore perform training in a low-rank subspace to reduce state memory, while recent work on GoLore provides robust subspace optimization schemes with convergence guarantees under stochastic noise [10]. Related subspace algorithms

for federated learning suggest that structured projections can be leveraged to cope with data heterogeneity while controlling complexity [41]. These ideas connect to geometric convergence theory for preconditioned iterations [18] and have been applied in FL with local matrix orthogonalization [17]. Strategies to reduce activation and optimizer-state overhead include gradient checkpointing and parameter-efficient fine-tuning, while architectural innovations like the efficient mixture-of-experts design in DeepSeek-V2 also contribute to system economy [6]. A critical challenge is the integrated co-design of memory and communication efficiency. This can involve compressing optimizer states, leveraging asynchronous decentralized training to reduce synchronization peaks [16], and employing dynamic topology management. Algorithms like AL-DSGD weight neighbors based on performance [7], while methods like TELEPORTATION activate only a subset of nodes [26]. The theoretical limits of such integrated systems, particularly under joint memory and communication constraints, remain an open area, highlighting the need for holistic designs that balance resource efficiency with convergence guarantees in heterogeneous environments.

Against this backdrop, FedSLoP sits at the intersection of federated optimization, communication compression, and memory-efficient training. By constraining stochastic gradients and momentum updates to random low-dimensional subspaces while preserving the standard FedAvg workflow, it directly targets the optimizer-state memory footprint on clients without introducing complex control variates or personalized heads. Our theoretical analysis builds on ideas from subspace optimization and compressed distributed SGD, but tailors them to the specific structure of random Stiefel projectors in FL, while our experiments connect these guarantees to practical gains in communication and memory usage on non-IID MNIST.

3 Method

Building on the problem formulation and motivation in Section 1, we now present the FedSLoP algorithm in detail. We first formalize the federated optimization problem and introduce the random low-rank subspace model used by our method, then describe the server and client updates, and finally discuss the memory and communication implications of the design.

3.1 Federated Optimization with Random Subspaces

We consider a standard cross-device federated learning setup with N clients. The global optimization problem is

$$\min_{\theta \in \mathbb{R}^d} f(\theta) := \frac{1}{N} \sum_{i=1}^N F_i(\theta), \quad (3.1)$$

where F_i is the local empirical risk on client i . At each communication round t , the server maintains a global model $\theta^t \in \mathbb{R}^d$ and coordinates local updates executed on a subset (or

all) of the clients. Our analysis in Section 4 focuses on the full-participation case, where all N clients participate in every communication round; extending the theory to partial participation is an interesting direction for future work.

A key ingredient of FedSLoP is a random low-rank subspace model applied to stochastic gradients and momentum. Let $r \leq d$ be a target rank, and define the Stiefel manifold

$$\text{St}(d, r) := \{P \in \mathbb{R}^{d \times r} : P^\top P = I_r\},$$

whose elements are column-orthonormal matrices. At the beginning of each communication round t , the server samples a random matrix $P_t \in \text{St}(d, r)$ from the uniform (Haar) distribution and forms the rank- r projector

$$\Pi_t := P_t P_t^\top \in \mathbb{R}^{d \times d}. \quad (3.2)$$

We denote the subspace ratio by

$$\underline{\delta} := \frac{r}{d} \in (0, 1], \quad (3.3)$$

which will play a central role in the convergence analysis in Section 4.

Intuitively, Π_t selects a different random r -dimensional subspace at each communication round and restricts all local optimization steps to this subspace, while the model itself remains full-dimensional. When $r = d$, the projector reduces to the identity and FedSLoP collapses to standard FedAvg with momentum; when $r \ll d$, the updates are effectively low-rank, dramatically reducing the dimensionality of the optimizer state and communicated directions.

3.2 FedSLoP with Full Participation

Algorithm 1 summarizes the core variant of FedSLoP analyzed in Section 4, specialized to full client participation and per-round zero-initialized momentum.

The algorithm follows the FedAvg template: the server repeatedly broadcasts the current global model, clients perform several local stochastic-gradient steps, and the server averages the resulting model deltas. The main difference is the use of the random projector Π_t inside the local momentum update, which constrains the effective update directions to a low-dimensional subspace shared across all clients within round t . This design enables memory-efficient implementations in which only projected gradients and momentum need to be stored and communicated, while the full model parameters reside in standard dense tensors.

3.3 Memory and Communication Considerations

The random subspace design in Algorithm 1 has several concrete implications for system resources:

Algorithm 1 FedSLoP with Full Participation

-
- 1: **Input:** number of rounds T , local steps τ , stepsize $\eta > 0$, momentum parameter $\mu \in [0, 1)$, rank r , number of clients N .
 - 2: **Server initializes:** global model $\theta^0 \in \mathbb{R}^d$.
 - 3: **for** each communication round $t = 0, 1, \dots, T-1$ **do**
 - 4: Sample $P_t \sim \text{Unif}(\text{St}(d, r))$ and set $\Pi_t = P_t P_t^\top$.
 - 5: Broadcast (θ^t, P_t) (or equivalently Π_t) to all clients $i \in \{1, \dots, N\}$.
 - 6: **for** each client $i = 1, \dots, N$ **in parallel do**
 - 7: $\Delta_i^t \leftarrow \text{CLIENTUPDATE}(i, \theta^t, \Pi_t)$.
 - 8: **end for**
 - 9: Aggregate updates and set

$$\theta^{t+1} \leftarrow \theta^t + \frac{1}{N} \sum_{i=1}^N \Delta_i^t.$$

- 10: **end for**
 - 11: **Output:** final global model θ^T .
 - 12: **procedure** CLIENTUPDATE(i, θ^t, Π_t)
 - 13: Initialize local model $\theta_{i,0} \leftarrow \theta^t$ and local momentum $v_{i,0} \leftarrow 0$.
 - 14: **for** local step $s = 0, 1, \dots, \tau-1$ **do**
 - 15: Sample a minibatch $\xi_{i,s}$ from client i .
 - 16: Compute stochastic gradient $g_{i,s} \leftarrow \nabla F_i(\theta_{i,s}; \xi_{i,s})$.
 - 17: Update momentum (projected) $v_{i,s+1} \leftarrow \mu v_{i,s} + \Pi_t g_{i,s}$.
 - 18: Update local model $\theta_{i,s+1} \leftarrow \theta_{i,s} - \eta v_{i,s+1}$.
 - 19: **end for**
 - 20: **return** $\Delta_i^t \leftarrow \theta_{i,\tau} - \theta^t$.
-

Optimizer-state memory. In standard FedAvg with momentum, each client typically maintains a full-dimensional momentum buffer of size d , in addition to the model parameters themselves. In FedSLoP, the momentum vector $v_{i,s}$ is updated using projected gradients $\Pi_t g_{i,s}$, so it can be stored and manipulated in the same d -dimensional space but its effective degrees of freedom are restricted to the r -dimensional subspace spanned by P_t . In practical implementations that exploit this structure, clients can store the momentum in factored form $v_{i,s} = P_t w_{i,s}$ with $w_{i,s} \in \mathbb{R}^r$, reducing the optimizer-state memory from $\mathcal{O}(d)$ to $\mathcal{O}(r)$ per linear layer. When $r \ll d$, this yields substantial savings for deep models.

Communication cost. On the downlink, the server must transmit the projector information in addition to the model parameters. However, P_t can be generated from shared random seeds or low-precision parameters, and its cost is amortized across all clients

within a round. On the uplink, clients send full-model deltas Δ_i^t , but these deltas arise from low-rank projected updates, meaning that their information content is effectively confined to an r -dimensional subspace. In practice, this structure can be combined with standard compression (e.g., low-rank factorization per layer) to reduce transmitted bits without altering the theoretical analysis in Section 4.

Compatibility with standard FL workflows. When the rank is set to $r = d$, the projector Π_t equals the identity and Algorithm 1 reduces to FedAvg with client-side momentum. Thus FedSLoP can be viewed as a strict generalization of FedAvg that introduces a tunable subspace ratio $\underline{\delta}$ controlling the trade-off between memory/communication and convergence accuracy. This perspective is particularly useful when interpreting the convergence bounds in Theorem 4.1, where $\underline{\delta}$ appears explicitly in both the optimization and variance terms.

In the next section we formalize the assumptions underlying our analysis and develop a detailed nonconvex convergence theory for Algorithm 1.

4 Theory

In this section we develop a detailed nonconvex convergence analysis of FedSLoP in the full-participation setting described in Section 3. We first state the smoothness and stochastic-gradient assumptions, then introduce the random subspace model and key auxiliary quantities, and finally present the main convergence theorem together with the supporting lemmas and corollaries. All proofs are provided in full.

4.1 Assumptions and Random Subspace Model

We recall the standard smoothness and bounded-variance assumptions.

Assumption 1 (Smoothness and lower boundedness). The global objective function f in (3.1) is bounded below by $f_* > -\infty$. The global function f and each local function F_i are L -smooth, i.e., their gradients are L -Lipschitz continuous.

Assumption 2 (Stochastic gradient properties). For each client i and model θ , the stochastic gradient $g_i(\theta)$ computed from a minibatch is an unbiased estimator of the true local gradient, $\mathbb{E}[g_i(\theta)] = \nabla F_i(\theta)$. The local variance is bounded as $\mathbb{E}[\|g_i(\theta) - \nabla F_i(\theta)\|^2] \leq \sigma_L^2$, and the gradient heterogeneity across clients is bounded as $\|\nabla F_i(\theta) - \nabla f(\theta)\|^2 \leq \sigma_G^2$.

In addition, FedSLoP uses the random projector Π_t defined in (3.2). Throughout this section we assume that P_t is drawn independently from the Haar measure on $\text{St}(d, r)$ at each communication round, independently of the data sampling and past randomness. As in Section 3, we denote the subspace ratio by $\underline{\delta} = r/d$.

We begin with a basic lemma on the properties of random Stiefel projectors.

Lemma 4.1 (Random Stiefel projection). *Let d, r be integers with $1 \leq r \leq d$, and let*

$$\text{St}_{d,r} := \{P \in \mathbb{R}^{d \times r} : P^\top P = I_r\}$$

be the (column) Stiefel manifold. Suppose P is sampled from the uniform (Haar) distribution on $\text{St}_{d,r}$. Then for any fixed vector $g \in \mathbb{R}^d$,

- (1) $\mathbb{E}[PP^\top] = \frac{r}{d} I_d$;
- (2) $\mathbb{E}[\|PP^\top g\|^2] = \frac{r}{d} \|g\|^2$;
- (3) $\mathbb{E}[\|(I_d - PP^\top)g\|^2] = (1 - \frac{r}{d}) \|g\|^2$.

Moreover, the same identities hold if we replace g by any fixed matrix $G \in \mathbb{R}^{d \times m}$ and interpret $\|\cdot\|$ as the Frobenius norm:

$$\mathbb{E}[\|PP^\top G\|_F^2] = \frac{r}{d} \|G\|_F^2, \quad \mathbb{E}[\|(I_d - PP^\top)G\|_F^2] = \left(1 - \frac{r}{d}\right) \|G\|_F^2.$$

Proof. Throughout, expectations are taken with respect to the Haar (uniform) distribution on $\text{St}_{d,r}$. By definition of this distribution, for every orthogonal matrix $Q \in O(d)$, the random matrices P and QP have the same distribution. In particular, for any measurable function f ,

$$\mathbb{E}[f(P)] = \mathbb{E}[f(QP)].$$

Step 1: Computation of $\mathbb{E}[PP^\top]$. Define the random projection matrix

$$\Pi := PP^\top \in \mathbb{R}^{d \times d}.$$

Note that Π is symmetric and idempotent:

$$\Pi^\top = \Pi, \quad \Pi^2 = PP^\top PP^\top = P(P^\top P)P^\top = P I_r P^\top = \Pi.$$

Let

$$M := \mathbb{E}[\Pi] = \mathbb{E}[PP^\top] \in \mathbb{R}^{d \times d}.$$

Fix any orthogonal $Q \in O(d)$. Using the distributional identity $P \stackrel{d}{=} QP$ and the fact that Q is deterministic, we obtain

$$M = \mathbb{E}[PP^\top] = \mathbb{E}[(QP)(QP)^\top] = \mathbb{E}[QPP^\top Q^\top] = Q\mathbb{E}[PP^\top]Q^\top = QMQ^\top.$$

Thus M satisfies

$$QMQ^\top = M, \quad \forall Q \in O(d). \quad (4.1)$$

By a standard representation-theoretic argument, the only matrices commuting with all orthogonal matrices are scalar multiples of the identity, so there exists $\lambda \in \mathbb{R}$ such that

$$M = \lambda I_d.$$

Taking traces and using $\text{tr}(PP^\top) = r$ almost surely, we obtain

$$\text{tr}(M) = \mathbb{E}[\text{tr}(PP^\top)] = r.$$

On the other hand, $\text{tr}(M) = \text{tr}(\lambda I_d) = \lambda d$, so $\lambda = r/d$. Hence

$$\mathbb{E}[PP^\top] = M = \frac{r}{d} I_d,$$

proving item (1).

Step 2: Action on a fixed vector. Let $g \in \mathbb{R}^d$ be fixed. Then

$$\mathbb{E}[\|PP^\top g\|^2] = \mathbb{E}[g^\top PP^\top PP^\top g] = \mathbb{E}[g^\top PP^\top g] = g^\top \mathbb{E}[PP^\top] g = \frac{r}{d} \|g\|^2,$$

which is item (2). Similarly,

$$\mathbb{E}[\|(I_d - PP^\top)g\|^2] = g^\top \mathbb{E}[I_d - PP^\top] g = g^\top \left(I_d - \frac{r}{d} I_d\right) g = \left(1 - \frac{r}{d}\right) \|g\|^2,$$

which is item (3).

Step 3: Matrix case and Frobenius norm. Let $G \in \mathbb{R}^{d \times m}$ be fixed, and write its columns as

$$G = [g_1 \dots g_m], \quad g_j \in \mathbb{R}^d.$$

Then

$$PP^\top G = [PP^\top g_1 \dots PP^\top g_m], \quad (I_d - PP^\top)G = [(I_d - PP^\top)g_1 \dots (I_d - PP^\top)g_m].$$

Recall that the Frobenius norm satisfies

$$\|G\|_F^2 = \sum_{j=1}^m \|g_j\|^2.$$

Therefore,

$$\|PP^\top G\|_F^2 = \sum_{j=1}^m \|PP^\top g_j\|^2, \quad \|(I_d - PP^\top)G\|_F^2 = \sum_{j=1}^m \|(I_d - PP^\top)g_j\|^2.$$

Taking expectations and applying the vector identities from Step 2 column-wise,

$$\mathbb{E}[\|PP^\top G\|_F^2] = \sum_{j=1}^m \mathbb{E}[\|PP^\top g_j\|^2] = \sum_{j=1}^m \frac{r}{d} \|g_j\|^2 = \frac{r}{d} \|G\|_F^2,$$

and

$$\mathbb{E}[\|(I_d - PP^\top)G\|_F^2] = \sum_{j=1}^m \mathbb{E}[\|(I_d - PP^\top)g_j\|^2] = \sum_{j=1}^m \left(1 - \frac{r}{d}\right) \|g_j\|^2 = \left(1 - \frac{r}{d}\right) \|G\|_F^2.$$

This yields the claimed Frobenius-norm identities and completes the proof. $\square$

4.2 Drift and One-Round Descent

The main technical challenge in analyzing FedSLoP is the interaction between local momentum, multiple local steps, and the random subspace constraint. Following the detailed derivations in the proof file, we introduce the per-round drift quantity

$$D_t := \frac{1}{N} \sum_{i=1}^N \sum_{s=0}^{\tau-1} \mathbb{E}_t [\|\theta_{i,s} - \theta^t\|^2], \quad (4.2)$$

where $\mathbb{E}_t[\cdot]$ denotes conditional expectation given θ^t and the randomness up to the beginning of round t . This quantity measures how far, on average, the local iterates drift away from the current global model within a communication round.

The following lemma provides an upper bound of the drift quantity.

Lemma 4.2 (Drift bound under projected momentum). *Suppose Assumptions 1 and 2 hold and the stepsize η satisfies the explicit condition*

$$\eta^2 \leq \frac{1-\mu^2}{6L^2\tau^3}, \quad (4.3)$$

it holds that

$$D_t \leq \frac{6\tau^4}{1-\mu^2} \eta^2 \left(\delta \|\nabla f(\theta^t)\|^2 + \sigma_L^2 + \sigma_G^2 \right). \quad (4.4)$$

Proof. We proceed in three steps.

Step 1: Expressing the drift in terms of momenta. Fix a round t and a client i . Under the per-round zero-momentum setup, the local recursions are

$$\begin{aligned} v_{i,s+1} &= \mu v_{i,s} + \Pi_t g_{i,s}, \\ \theta_{i,s+1} &= \theta_{i,s} - \eta v_{i,s+1}, \quad s=0, \dots, \tau-1, \end{aligned}$$

with $\theta_{i,0} = \theta^t$ and $v_{i,0} = 0$. By telescoping, for any $s \geq 1$,

$$\theta_{i,s} - \theta^t = -\eta \sum_{u=0}^{s-1} v_{i,u+1}.$$

Hence, by Cauchy–Schwarz and the bound $s \leq \tau$,

$$\|\theta_{i,s} - \theta^t\|^2 = \eta^2 \left\| \sum_{u=0}^{s-1} v_{i,u+1} \right\|^2 \leq \eta^2 s \sum_{u=0}^{s-1} \|v_{i,u+1}\|^2 \leq \eta^2 \tau \sum_{u=0}^{\tau-1} \|v_{i,u+1}\|^2.$$

Summing over $s=0, \dots, \tau-1$ (the $s=0$ term is zero) yields

$$\sum_{s=0}^{\tau-1} \|\theta_{i,s} - \theta^t\|^2 \leq \eta^2 \tau^2 \sum_{u=0}^{\tau-1} \|v_{i,u+1}\|^2.$$

Averaging over clients and taking conditional expectation gives

$$D_t = \frac{1}{N} \sum_{i=1}^N \sum_{s=0}^{\tau-1} \mathbb{E}_t [\|\theta_{i,s} - \theta^t\|^2] \leq \eta^2 \tau^2 \frac{1}{N} \sum_{i=1}^N \sum_{u=0}^{\tau-1} \mathbb{E}_t [\|v_{i,u+1}\|^2]. \quad (4.5)$$

Step 2: Bounding the momentum norms. Unrolling the momentum recursion gives, for any $u \geq 0$,

$$v_{i,u+1} = \mu^{u+1} v_{i,0} + \sum_{q=0}^u \mu^{u-q} \Pi_t g_{i,q}.$$

Using $v_{i,0} = 0$, this simplifies to

$$v_{i,u+1} = \sum_{q=0}^u \mu^{u-q} \Pi_t g_{i,q}.$$

Let $a_q := \mu^{u-q} \Pi_t g_{i,q}$. By Cauchy-Schwarz,

$$\left\| \sum_{q=0}^u a_q \right\|^2 \leq (u+1) \sum_{q=0}^u \|a_q\|^2 \leq \tau \sum_{q=0}^u \mu^{2(u-q)} \|\Pi_t g_{i,q}\|^2,$$

where we used $u+1 \leq \tau$. Thus

$$\|v_{i,u+1}\|^2 \leq \tau \sum_{q=0}^u \mu^{2(u-q)} \|\Pi_t g_{i,q}\|^2.$$

Summing over $u = 0, \dots, \tau-1$ and rearranging the finite sums yields

$$\begin{aligned} \sum_{u=0}^{\tau-1} \|v_{i,u+1}\|^2 &\leq \tau \sum_{u=0}^{\tau-1} \sum_{q=0}^u \mu^{2(u-q)} \|\Pi_t g_{i,q}\|^2 \\ &= \tau \sum_{q=0}^{\tau-1} \left(\sum_{u=q}^{\tau-1} \mu^{2(u-q)} \right) \|\Pi_t g_{i,q}\|^2 \\ &\leq \frac{\tau}{1-\mu^2} \sum_{q=0}^{\tau-1} \|\Pi_t g_{i,q}\|^2, \end{aligned}$$

where we used the geometric-series bound $\sum_{u=q}^{\tau-1} \mu^{2(u-q)} \leq \sum_{k=0}^{\infty} \mu^{2k} = 1/(1-\mu^2)$. Substituting into (4.5), we obtain

$$\begin{aligned} D_t &\leq \eta^2 \tau^2 \frac{1}{N} \sum_{i=1}^N \mathbb{E}_t \left[\sum_{u=0}^{\tau-1} \|v_{i,u+1}\|^2 \right] \\ &\leq \eta^2 \tau^2 \frac{1}{N} \sum_{i=1}^N \mathbb{E}_t \left[\frac{\tau}{1-\mu^2} \sum_{q=0}^{\tau-1} \|\Pi_t g_{i,q}\|^2 \right] \\ &= \frac{\eta^2 \tau^3}{1-\mu^2} \frac{1}{N} \sum_{i=1}^N \sum_{q=0}^{\tau-1} \mathbb{E}_t [\|\Pi_t g_{i,q}\|^2]. \end{aligned} \tag{4.6}$$

Step 3: Bounding $\mathbb{E}_t[\|\Pi_t g_{i,q}\|^2]$ and closing the inequality. By definition $g_{i,q} = \nabla F_i(\theta_{i,q}; \xi_{i,q})$. Using Assumption 2, we have

$$\begin{aligned} \mathbb{E}_t[\|\Pi_t g_{i,q}\|^2] &= \mathbb{E}_t[\|\Pi_t(\nabla F_i(\theta_{i,q}; \xi_{i,q}) - \nabla F_i(\theta_{i,q}))\|^2] + \mathbb{E}_t[\|\Pi_t \nabla F_i(\theta_{i,q})\|^2] \\ &\leq \mathbb{E}_t[\|(\nabla F_i(\theta_{i,q}; \xi_{i,q}) - \nabla F_i(\theta_{i,q}))\|^2] + \mathbb{E}_t[\|\Pi_t \nabla F_i(\theta_{i,q})\|^2] \\ &\leq \sigma_L^2 + \mathbb{E}_t[\|\Pi_t \nabla F_i(\theta_{i,q})\|^2]. \end{aligned} \quad (4.7)$$

Next we bound $\|\Pi_t \nabla F_i(\theta_{i,q})\|^2$ using Assumptions 1 and 2. Insert and subtract $\Pi_t \nabla F_i(\theta^t)$ and $\Pi_t \nabla f(\theta^t)$:

$$\Pi_t \nabla F_i(\theta_{i,q}) = (\Pi_t \nabla F_i(\theta_{i,q}) - \Pi_t \nabla F_i(\theta^t)) + (\Pi_t \nabla F_i(\theta^t) - \Pi_t \nabla f(\theta^t)) + \Pi_t \nabla f(\theta^t).$$

Using $\|a+b+c\|^2 \leq 3(\|a\|^2 + \|b\|^2 + \|c\|^2)$, the L -smoothness of F_i and the global variance bound in Assumption 2, we obtain

$$\begin{aligned} \|\Pi_t \nabla F_i(\theta_{i,q})\|^2 &\leq 3\|\Pi_t \nabla F_i(\theta_{i,q}) - \Pi_t \nabla F_i(\theta^t)\|^2 + 3\|\Pi_t \nabla F_i(\theta^t) - \Pi_t \nabla f(\theta^t)\|^2 + 3\|\Pi_t \nabla f(\theta^t)\|^2 \\ &\leq 3\|\nabla F_i(\theta_{i,q}) - \nabla F_i(\theta^t)\|^2 + 3\|\nabla F_i(\theta^t) - \nabla f(\theta^t)\|^2 + 3\|\Pi_t \nabla f(\theta^t)\|^2 \\ &\leq 3L^2\|\theta_{i,q} - \theta^t\|^2 + 3\sigma_G^2 + 3\|\Pi_t \nabla f(\theta^t)\|^2. \end{aligned}$$

Taking conditional expectation and applying Lemma 4.1 achieves

$$\mathbb{E}_t[\|\Pi_t \nabla F_i(\theta_{i,q})\|^2] \leq 3L^2\mathbb{E}_t[\|\theta_{i,q} - \theta^t\|^2] + 3\sigma_G^2 + 3\underline{\delta}\|\nabla f(\theta^t)\|^2. \quad (4.8)$$

Combining (4.7) and (4.8) gives

$$\mathbb{E}_t[\|\Pi_t g_{i,q}\|^2] \leq \sigma_L^2 + 3\underline{\delta}\|\nabla f(\theta^t)\|^2 + 3L^2\mathbb{E}_t[\|\theta_{i,q} - \theta^t\|^2] + 3\sigma_G^2. \quad (4.9)$$

Substituting (4.9) into (4.6), we obtain

$$\begin{aligned} D_t &\leq \frac{\eta^2 \tau^3}{1-\mu^2} \frac{1}{N} \sum_{i=1}^N \sum_{q=0}^{\tau-1} \left(3\underline{\delta}\|\nabla f(\theta^t)\|^2 + 3L^2\mathbb{E}_t[\|\theta_{i,q} - \theta^t\|^2] + \sigma_L^2 + 3\sigma_G^2 \right) \\ &= \frac{\eta^2 \tau^3}{1-\mu^2} \left(3\underline{\delta}\|\nabla f(\theta^t)\|^2 \cdot \tau + 3L^2 D_t + (\sigma_L^2 + 3\sigma_G^2) \cdot \tau \right) \\ &= \frac{3\eta^2 \tau^4 \underline{\delta}}{1-\mu^2} \|\nabla f(\theta^t)\|^2 + \frac{\eta^2 \tau^4}{1-\mu^2} (\sigma_L^2 + 3\sigma_G^2) + \frac{3L^2 \eta^2 \tau^3}{1-\mu^2} D_t. \end{aligned} \quad (4.10)$$

Rearranging (4.10) gives

$$\left(1 - \frac{3L^2 \eta^2 \tau^3}{1-\mu^2}\right) D_t \leq \frac{3\eta^2 \tau^4 \underline{\delta}}{1-\mu^2} \|\nabla f(\theta^t)\|^2 + \frac{\eta^2 \tau^4}{1-\mu^2} (\sigma_L^2 + 3\sigma_G^2).$$

If the stepsize satisfies (4.3), i.e.

$$\eta^2 \leq \frac{1-\mu^2}{6L^2\tau^3},$$

then $3L^2\eta^2\tau^3/(1-\mu^2) \leq 1/2$ and therefore

$$1 - \frac{3L^2\eta^2\tau^3}{1-\mu^2} \geq \frac{1}{2}.$$

This implies

$$\begin{aligned} D_t &\leq \frac{6\eta^2\tau^4\delta}{1-\mu^2} \|\nabla f(\theta^t)\|^2 + \frac{2\eta^2\tau^4}{1-\mu^2} (\sigma_L^2 + 3\sigma_G^2) \\ &\leq \frac{6\tau^4}{1-\mu^2} \eta^2 \left(\delta \|\nabla f(\theta^t)\|^2 + \sigma_L^2 + \sigma_G^2 \right), \end{aligned}$$

which is (4.4). $\square$

The next ingredient is a one-round descent inequality that combines smoothness, the random projector, and the drift quantity.

Lemma 4.3 (One-round descent under random projector). *Under Assumptions 1 and 2, if $\eta \leq 1/(LS_\tau)$, the random subspace one-round descent satisfies*

$$\mathbb{E}_t[f(\theta^{t+1})] \leq f(\theta^t) - \frac{\delta}{2}\eta S_\tau \|\nabla f(\theta^t)\|^2 + \frac{\eta^2 S_\tau^2 L \sigma_L^2}{N} + \frac{2L^2\eta}{\delta(1-\mu)} D_t, \quad (4.11)$$

where $S_\tau = \sum_{q=0}^{\tau-1} \alpha_{\tau,q}$ and $\alpha_{\tau,q} = (1 - \mu^{\tau-q}) / (1 - \mu)$ are the effective momentum weights.

Proof. We fix a communication round t and work with the conditional expectation $\mathbb{E}_t[\cdot]$ given θ^t and all past randomness. Recall that the server update can be written as

$$\theta^{t+1} = \theta^t + \bar{\Delta}_t, \quad \bar{\Delta}_t := \frac{1}{N} \sum_{i=1}^N (\theta_{i,\tau} - \theta^t).$$

We first express $\bar{\Delta}_t$ in terms of the projected local gradients, then bound the linear descent term and the quadratic remainder term separately, and finally combine the bounds.

Step 1: Expressing the aggregated update. From the local recursions,

$$v_{i,s+1} = \mu v_{i,s} + \Pi_i g_{i,s}, \quad \theta_{i,s+1} = \theta_{i,s} - \eta v_{i,s+1}, \quad s = 0, \dots, \tau-1,$$

with $v_{i,0} = 0$ and $\theta_{i,0} = \theta^t$, it follows by telescoping that

$$\theta_{i,\tau} - \theta^t = -\eta \sum_{s=0}^{\tau-1} v_{i,s+1}.$$

Unrolling the momentum recursion gives, for every $u \geq 0$,

$$v_{i,u+1} = \sum_{q=0}^u \mu^{u-q} \Pi_t g_{i,q},$$

and hence

$$\begin{aligned} \theta_{i,\tau} - \theta^t &= -\eta \sum_{s=0}^{\tau-1} v_{i,s+1} = -\eta \sum_{s=0}^{\tau-1} \sum_{q=0}^s \mu^{s-q} \Pi_t g_{i,q} \\ &= -\eta \sum_{q=0}^{\tau-1} \left(\sum_{s=q}^{\tau-1} \mu^{s-q} \right) \Pi_t g_{i,q} = -\eta \sum_{q=0}^{\tau-1} \alpha_{\tau,q} \Pi_t g_{i,q}, \end{aligned}$$

where $\alpha_{\tau,q} = (1 - \mu^{\tau-q}) / (1 - \mu)$ as in the lemma statement. Averaging over clients, we obtain

$$\bar{\Delta}_t = -\eta \sum_{q=0}^{\tau-1} \alpha_{\tau,q} \Pi_t \bar{g}_q, \quad \bar{g}_q := \frac{1}{N} \sum_{i=1}^N g_{i,q}. \quad (4.12)$$

Step 2: Bounding the linear descent term. By L -smoothness of f we have the standard one-step inequality

$$f(\theta^{t+1}) \leq f(\theta^t) + \langle \nabla f(\theta^t), \bar{\Delta}_t \rangle + \frac{L}{2} \|\bar{\Delta}_t\|^2.$$

Taking $\mathbb{E}_t[\cdot]$ and substituting (4.12),

$$\mathbb{E}_t[f(\theta^{t+1})] \leq f(\theta^t) - \eta \sum_{q=0}^{\tau-1} \alpha_{\tau,q} \mathbb{E}_t[\langle \nabla f(\theta^t), \Pi_t \bar{g}_q \rangle] + \frac{L}{2} \mathbb{E}_t[\|\bar{\Delta}_t\|^2]. \quad (4.13)$$

Fix a local step index q and set $g_* := \nabla f(\theta^t)$. We decompose

$$\langle g_*, \Pi_t \bar{g}_q \rangle = \langle g_*, \Pi_t g_* \rangle + \langle g_*, \Pi_t (\bar{g}_q - g_*) \rangle.$$

Since g_* depends only on θ^t and the projector Π_t is sampled independently from the Stiefel manifold, Lemma 4.1 (random Stiefel projection) yields

$$\mathbb{E}_t[\langle g_*, \Pi_t g_* \rangle] = \underline{\delta} \|g_*\|^2.$$

For the second term we use Assumption 2, Cauchy–Schwarz, the fact that Π_t is a contraction, and Young’s inequality with parameter $\underline{\delta}/2$:

$$\begin{aligned} |\mathbb{E}_t[\langle g_*, \Pi_t (\bar{g}_q - g_*) \rangle]| &= |\mathbb{E}_t[\langle g_*, \Pi_t (\tilde{g}_q - g_*) \rangle]| \\ &\leq \|g_*\| \mathbb{E}_t[\|\tilde{g}_q - g_*\|] \\ &\leq \frac{\underline{\delta}}{4} \|g_*\|^2 + \frac{1}{\underline{\delta}} \mathbb{E}_t[\|\tilde{g}_q - g_*\|^2], \end{aligned}$$

where we define

$$\tilde{g}_q := \frac{1}{N} \sum_{i=1}^N \nabla F_i(\theta_{i,q}).$$

Combining the two pieces gives

$$\mathbb{E}_t [\langle g_*, \Pi_t \tilde{g}_q \rangle] \geq \frac{3\delta}{4} \|\nabla f(\theta^t)\|^2 - \frac{1}{\underline{\delta}} \mathbb{E}_t [\|\tilde{g}_q - \nabla f(\theta^t)\|^2]. \quad (4.14)$$

Substituting (4.14) into (4.13), we obtain

$$\mathbb{E}_t [f(\theta^{t+1})] \leq f(\theta^t) - \frac{3\delta}{4} \eta S_\tau \|\nabla f(\theta^t)\|^2 + \frac{\eta}{\underline{\delta}} \sum_{q=0}^{\tau-1} \alpha_{\tau,q} \mathbb{E}_t [\|\tilde{g}_q - \nabla f(\theta^t)\|^2] + \frac{L}{2} \mathbb{E}_t [\|\bar{\Delta}_t\|^2]. \quad (4.15)$$

Step 3: Controlling the stochastic-gradient variance and drift. We now bound the variance term in (4.15) and the quadratic term $\mathbb{E}_t \|\bar{\Delta}_t\|^2$. Using L -smoothness of F_i and Jensen's inequality, we have

$$\begin{aligned} \mathbb{E}_t [\|\tilde{g}_q - \nabla f(\theta^t)\|^2] &= \mathbb{E}_t \left[\left\| \frac{1}{N} \sum_{i=1}^N (\nabla F_i(\theta_{i,q}) - \nabla F_i(\theta^t)) \right\|^2 \right] \\ &\leq \frac{1}{N} \sum_{i=1}^N \mathbb{E}_t \left[\left\| \nabla F_i(\theta_{i,q}) - \nabla F_i(\theta^t) \right\|^2 \right] \\ &\leq L^2 \bar{D}_q, \end{aligned} \quad (4.16)$$

where we define

$$\bar{D}_q := \frac{1}{N} \sum_{i=1}^N \mathbb{E}_t [\|\theta_{i,q} - \theta^t\|^2].$$

Using $\alpha_{\tau,q} \leq 1/(1-\mu)$, we obtain

$$\frac{\eta}{\underline{\delta}} \sum_{q=0}^{\tau-1} \alpha_{\tau,q} \mathbb{E}_t [\|\tilde{g}_q - \nabla f(\theta^t)\|^2] \leq \frac{L^2 \eta D_t}{\underline{\delta}(1-\mu)}. \quad (4.17)$$

Write

$$g_{i,q} = \nabla F_i(\theta_{i,q}; \xi_{i,q}) = \nabla F_i(\theta_{i,q}) + (g_{i,q} - \nabla F_i(\theta_{i,q})),$$

so that $\tilde{g}_q - \nabla f(\theta^t)$ can be decomposed into (i) stochastic-noise fluctuations around the local true gradients and (ii) the bias between local and global gradients induced by data heterogeneity and local drift. A standard variance decomposition using Assumption 2

(unbiasedness and bounded local variance) and the bounded heterogeneity condition gives

$$\begin{aligned}\mathbb{E}_t[\|\bar{g}_q - \nabla f(\theta^t)\|^2] &= \mathbb{E}_t\left[\left\|\frac{1}{N}\sum_{i=1}^N(\nabla F_i(\theta_{i,q}) - \nabla F_i(\theta^t))\right\|^2\right] + \sum_{i=1}^N \mathbb{E}_t\left[\left\|\frac{1}{N}(\bar{g}_{i,q} - \nabla F_i(\theta_{i,q}))\right\|^2\right] \\ &\leq \frac{1}{N}\sum_{i=1}^N \mathbb{E}_t[\|\nabla F_i(\theta_{i,q}) - \nabla F_i(\theta^t)\|^2] + \frac{\sigma_L^2}{N} \\ &\leq L^2 \bar{D}_q + \frac{\sigma_L^2}{N},\end{aligned}\tag{4.18}$$

where we have used the L -smoothness of F_i to control $\|\nabla F_i(\theta_{i,q}) - \nabla F_i(\theta^t)\|^2$ by $L^2 \mathbb{E}_t\|\theta_{i,q} - \theta^t\|^2$.

Next, from the expansion (4.12) and the inequality $\|\sum_q a_q x_q\|^2 \leq (\sum_q a_q) \sum_q a_q \|x_q\|^2$ for $a_q \geq 0$, we have

$$\begin{aligned}\mathbb{E}_t[\|\bar{\Delta}_t\|^2] &\leq \eta^2 S_\tau \sum_{q=0}^{\tau-1} \alpha_{\tau,q} \mathbb{E}_t[\|\Pi_t \bar{g}_q\|^2] \\ &\leq 2\eta^2 S_\tau \sum_{q=0}^{\tau-1} \alpha_{\tau,q} (\mathbb{E}_t[\|\Pi_t \nabla f(\theta^t)\|^2] + \mathbb{E}_t[\|\bar{g}_q - \nabla f(\theta^t)\|^2]) \\ &\leq 2\eta^2 S_\tau \left(\underline{\delta} S_\tau \|\nabla f(\theta^t)\|^2 + S_\tau \frac{\sigma_L^2}{N} + \frac{L^2}{1-\mu} D_t \right),\end{aligned}\tag{4.19}$$

where we used the bound $\|\Pi_t \bar{g}_q\|^2 \leq 2\|\Pi_t \nabla f(\theta^t)\|^2 + 2\|\Pi_t(\bar{g}_q - \nabla f(\theta^t))\|^2$, (4.18), Lemma 4.1 and $\alpha_{\tau,q} \leq 1/(1-\mu)$.

Step 4: Collecting terms. Plugging (4.17) and (4.19) into (4.15) and absorbing numerical constants, we arrive at

$$\mathbb{E}_t[f(\theta^{t+1})] \leq f(\theta^t) - \frac{\delta}{4} \eta S_\tau (3 - 4L\eta S_\tau) \|\nabla f(\theta^t)\|^2 + \eta^2 S_\tau^2 L \frac{\sigma_L^2}{N} + \left(\frac{L^2 \eta}{\underline{\delta}(1-\mu)} + \frac{L^3 \eta^2 S_\tau}{1-\mu} \right) D_t.$$

When $\eta \leq 1/(LS_\tau)$, we further have

$$\mathbb{E}_t[f(\theta^{t+1})] \leq f(\theta^t) - \frac{\delta}{2} \eta S_\tau \|\nabla f(\theta^t)\|^2 + \frac{\eta^2 S_\tau^2 L \sigma_L^2}{N} + \frac{2L^2 \eta}{\underline{\delta}(1-\mu)} D_t,$$

which is exactly the claimed one-round descent inequality (4.11). This completes the proof. $\square$

Combining Lemmas 4.2 and 4.3 yields a clean one-round bound with explicit constants.

Lemma 4.4 (One-round descent with drift control). *Suppose Assumptions 1 and 2 hold, and that the stepsize η satisfies*

$$\eta \leq \min \left\{ \sqrt{\frac{1-\mu^2}{6L^2\tau^3}}, \frac{1}{LS_\tau} \right\}. \quad (4.20)$$

Then for Algorithm 1 we have, for every round t ,

$$\begin{aligned} \mathbb{E}_t[f(\theta^{t+1})] &\leq f(\theta^t) - \left(\frac{\delta}{2}\eta S_\tau - \frac{12L^2\tau^4\eta^3}{(1-\mu)(1-\mu^2)} \right) \|\nabla f(\theta^t)\|^2 \\ &\quad + \frac{\eta^2 S_\tau^2 L \sigma_L^2}{N} + \frac{12L^2\tau^4\eta^3}{\delta(1-\mu)(1-\mu^2)} (\sigma_L^2 + \sigma_G^2). \end{aligned} \quad (4.21)$$

Proof. Substituting the drift bound (4.4) from Lemma 4.2 into the one-round descent (4.11), and collecting the terms in $\|\nabla f(\theta^t)\|^2$ and $(\sigma_L^2 + \sigma_G^2)$, yields (4.21) with the stated constants. $\square$

4.3 Main Full-Participation Convergence Theorem

We are now ready to state and prove the main convergence theorem for FedSLoP.

Theorem 4.1 (Convergence of FedSLoP with explicit constants). *Suppose Assumptions 1 and 2 hold, and the algorithm is executed as described in Algorithm 1.*

For the momentum weights $\alpha_{\tau,q} = (1 - \mu^{\tau-q}) / (1 - \mu)$ and $S_\tau = \sum_{q=0}^{\tau-1} \alpha_{\tau,q}$ defined above and the rank ratio $\delta = r/d \in (0,1]$, assume that the stepsize $\eta > 0$ obeys the explicit conditions

$$\eta \leq \min \left\{ \sqrt{\frac{1-\mu^2}{6L^2\tau^3}}, \frac{1}{LS_\tau}, \sqrt{\frac{\delta(1-\mu)(1-\mu^2)S_\tau}{48L^2\tau^4}} \right\}. \quad (4.22)$$

Then for every integer $T \geq 1$, the global iterates $(\theta^t)_{t \geq 0}$ of FedSLoP satisfy

$$\begin{aligned} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla f(\theta^t)\|^2] &\leq \frac{4}{\delta\eta S_\tau T} (f(\theta^0) - f_*) \\ &\quad + \frac{4\eta S_\tau L \sigma_L^2}{\delta N} + \frac{48\eta^2 \tau^4 L^2}{\delta^2(1-\mu)(1-\mu^2)S_\tau} (\sigma_L^2 + \sigma_G^2). \end{aligned} \quad (4.23)$$

Proof. We combine the explicit one-round descent inequality under random Stiefel projector with the explicit drift bound.

Step 1: One-round descent with drift. By Lemma 4.4, for each round t and conditional expectation $\mathbb{E}_t[\cdot]$ given θ^t we have

$$\begin{aligned} \mathbb{E}_t[f(\theta^{t+1})] &\leq f(\theta^t) - \left(\frac{\delta}{2}\eta S_\tau - \frac{12L^2\tau^4\eta^3}{(1-\mu)(1-\mu^2)} \right) \|\nabla f(\theta^t)\|^2 \\ &\quad + \frac{\eta^2 S_\tau^2 L \sigma_L^2}{N} + \frac{12L^2\tau^4\eta^3}{\delta(1-\mu)(1-\mu^2)} (\sigma_L^2 + \sigma_G^2). \end{aligned}$$

Step 2: Ensuring a negative gradient coefficient. Imposing the additional stepsize condition

$$\eta \leq \sqrt{\frac{\underline{\delta}(1-\mu)(1-\mu^2)S_\tau}{48L^2\tau^4}},$$

which is equivalent to

$$\frac{12L^2\tau^4}{(1-\mu)(1-\mu^2)}\eta^3 \leq \frac{\delta}{4}\eta S_\tau,$$

the coefficient in front of $\|\nabla f(\theta^t)\|^2$ is

$$-\frac{\delta}{2}\eta S_\tau + \frac{12L^2\tau^4}{(1-\mu)(1-\mu^2)}\eta^3 \leq -\frac{\delta}{4}\eta S_\tau,$$

so the one-round bound simplifies to

$$\begin{aligned} \mathbb{E}_t[f(\theta^{t+1})] &\leq f(\theta^t) - \frac{\delta}{4}\eta S_\tau \|\nabla f(\theta^t)\|^2 \\ &\quad + \frac{\eta^2 S_\tau^2 L \sigma_L^2}{N} + \frac{12L^2\tau^4\eta^3}{\underline{\delta}(1-\mu)(1-\mu^2)}(\sigma_L^2 + \sigma_G^2). \end{aligned} \quad (4.24)$$

This bound holds whenever (4.22) is satisfied.

Step 3: Telescoping and averaging. Taking full expectation in (4.24) and summing over $t=0, \dots, T-1$ yields

$$\begin{aligned} \mathbb{E}[f(\theta^T)] &\leq f(\theta^0) - \frac{\delta}{4}\eta S_\tau \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla f(\theta^t)\|^2] \\ &\quad + \frac{\eta^2 S_\tau^2 L T \sigma_L^2}{N} + \frac{12\eta^3 \tau^4 L^2 T}{\underline{\delta}(1-\mu)(1-\mu^2)}(\sigma_L^2 + \sigma_G^2). \end{aligned}$$

Using the lower boundedness $f(\theta^T) \geq f_*$, dividing both sides by T and by $\underline{\delta}\eta S_\tau/4$ and rearranging gives

$$\begin{aligned} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla f(\theta^t)\|^2] &\leq \frac{4}{\underline{\delta}\eta S_\tau T} (f(\theta^0) - f_*) \\ &\quad + \frac{4\eta S_\tau L \sigma_L^2}{\underline{\delta}N} + \frac{48\eta^2 \tau^4 L^2}{\underline{\delta}^2(1-\mu)(1-\mu^2)S_\tau} (\sigma_L^2 + \sigma_G^2), \end{aligned}$$

which is exactly the bound (4.23) stated in the theorem. $\square$

4.4 Asymptotic Rate Under Carefully Selected Step-Size

The explicit convergence bound in Theorem 4.1 can be optimized over the step-size η to reveal the asymptotic dependence on the number of rounds T and the problem parameters. The following corollary summarizes the result for a carefully selected step-size.

Corollary 4.1 (Optimal convergence rate via carefully-selected step-size). *Under the assumptions of Theorem 4.1, let $\Delta_0 := f(\theta^0) - f_*$. If we choose*

$$\eta = \left(\sqrt{\frac{6L^2\tau^3}{1-\mu^2}} + LS_\tau + \sqrt{\frac{48L^2\tau^4}{\underline{\delta}(1-\mu)(1-\mu^2)S_\tau}} + \sqrt{\frac{S_\tau^2LT\sigma_L^2}{N\Delta_0}} + \sqrt[3]{\frac{12\tau^4L^2T(\sigma_L^2+\sigma_G^2)}{\underline{\delta}(1-\mu)(1-\mu^2)\Delta_0}} \right)^{-1},$$

the averaged squared gradient norm satisfies

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} [\|\nabla f(\theta^t)\|^2] = \mathcal{O} \left(\underbrace{\frac{\Delta_0^{1/2}L^{1/2}\sigma_L}{\underline{\delta}N^{1/2}T^{1/2}}}_{\text{stochastic term}} + \underbrace{\frac{\Delta_0^{2/3}L^{2/3}\tau^{1/3}(\sigma_L^2+\sigma_G^2)^{1/3}}{\underline{\delta}^{4/3}T^{2/3}}}_{\text{higher-order drift term}} + \underbrace{\frac{\Delta_0L\tau^{1/2}}{\underline{\delta}T}}_{\text{optimization term}} \right).$$

where $\mathcal{O}(\cdot)$ hides only absolute numerical constants (independent of $L, \tau, \underline{\delta}, N, \sigma_L, \sigma_G$). In particular, the contribution of the local stochastic variance σ_L^2 enjoys a linear speedup factor $1/N$ in the dominant variance term.

Theorem 4.1 and Corollary 4.1 together provide a detailed nonconvex convergence theory for FedSLoP. The explicit dependence on the subspace ratio $\underline{\delta}$ clarifies how low-rank random projections affect the convergence property.

5 Numerical Experiments

We now evaluate FEDSLOP on a standard non-IID MNIST classification task and compare it against several federated baselines. The main goals are to assess convergence speed, final test accuracy, robustness to data heterogeneity, and the effect of the projection rank and the number of clients.

5.1 Experimental Setup

Dataset and task. We consider 10-class classification on MNIST. The training set contains 60,000 grayscale images and the test set contains 10,000 images. Each image is vectorized into $\mathbb{R}^{784}$. The prediction model is a two-layer multilayer perceptron with architecture $784 \rightarrow 128 \rightarrow 10$ and ReLU activation in the hidden layer. The loss function is the standard cross-entropy loss.

Table 1: Main federated configuration on non-IID MNIST.

Parameter	Value
Number of clients p	50
Dirichlet concentration α	0.1 (main), $\{0.05, 0.1, 0.5, 1.0\}$ (sensitivity)
Communication rounds T	100
Local epochs per round	1
Mini-batch size b	32
Learning rate η	0.01839685275139246
FEDSLOP projection rank r	112 (main), varied in ablation
FEDSLOP momentum	≈ 0.805
FEDLoRAWithMOMENTUM rank	15

Federated partition. We use 50 clients and distribute the training data by label-wise Dirichlet partitioning. In the main experiment the Dirichlet concentration parameter is $\alpha = 0.1$, which creates a strongly heterogeneous non-IID setting. In the heterogeneity study we vary $\alpha \in \{0.05, 0.1, 0.5, 1.0\}$.

Compared methods. We compare the proposed FEDSLOP against the following baselines:

- **FedAvgWithMomentum:** full-parameter federated averaging with server-side momentum;
- **FedMef:** magnitude-based sparse update baseline;
- **FederatedSelect:** random parameter-subset update baseline;
- **NeuLite:** block-wise partial-update baseline;
- **FedLoRAWithMomentum:** LoRA-based low-rank model adaptation with server-side momentum.

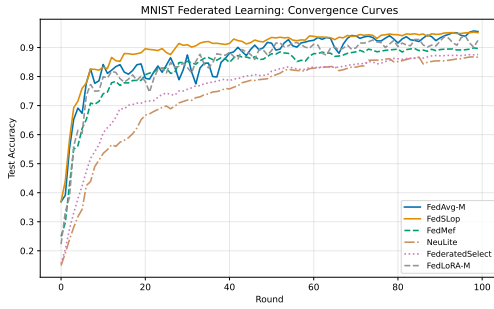
Training configuration. In the main comparison we train for $T = 100$ communication rounds, use one local epoch per round, batch size 32, and three random seeds $\{0, 1, 2\}$. The learning rate is fixed to

$$\eta = 0.01839685275139246,$$

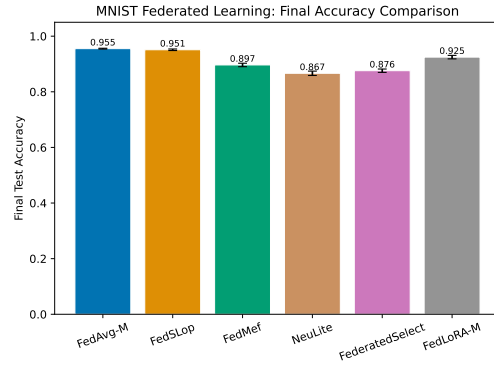
which is selected from the tuning procedure used for FEDSLOP and then reused across methods for a fair comparison. For the proposed method the main projection rank is $r = 112$. For the LoRA baseline we use rank 15.

Table 2: Final test accuracy on non-IID MNIST with $\alpha = 0.1$ after $T = 100$ communication rounds (mean $\pm$ standard deviation over three seeds).

Method	Final test accuracy (mean $\pm$ std)	Compression ratio
FedAvgWithMomentum	0.9551 ± 0.0015	1
FedSLoP	0.9511 ± 0.0028	15.3%
FedLoRAWithMomentum	0.9247 ± 0.0063	15.5%
FedMef	0.8965 ± 0.0062	15.0%
FederatedSelect	0.8759 ± 0.0060	15.0%
NeuLite	0.8665 ± 0.0078	15.0%



Test accuracy vs. communication rounds



Final test accuracy at round $T = 100$

Figure 1: Convergence and final precision on non-IID MNIST ($\alpha=0.1$). Left: test accuracy over communication rounds. Right: final test accuracy.

5.2 Main Comparison on Non-IID MNIST

Table 2 reports the final test accuracy after 100 communication rounds. The strongest baseline is FEDAVGWITHMOMENTUM, which reaches 0.9551 ± 0.0015 . The proposed FEDSLOP achieves 0.9511 ± 0.0028 , remaining very close to the full-parameter baseline while clearly outperforming the sparse and partial-update methods. The LoRA baseline also performs competitively, but still trails FEDSLOP by more than two percentage points.

Figure 1 shows the corresponding convergence curves and final-accuracy summary. The gap between FEDSLOP and FEDAVGWITHMOMENTUM stays small throughout training, whereas the sparse and blockwise baselines converge more slowly and saturate at lower accuracies. This indicates that low-rank projection of the optimizer state preserves most of the optimization benefit of full-parameter training while substantially reducing the effective update dimension.

Table 3: Final test accuracy on MNIST after $T=100$ communication rounds for different Dirichlet parameters α (mean over three seeds). Smaller α means stronger heterogeneity.

Method	$\alpha = 0.05$	$\alpha = 0.10$	$\alpha = 0.50$	$\alpha = 1.00$
FedAvgWithMomentum	0.9035	0.9551	0.9722	0.9739
FedSLoP	0.9269	0.9511	0.9700	0.9708
FedLoRAWithMomentum	0.8838	0.9247	0.9558	0.9616
FedMef	0.8709	0.8965	0.9283	0.9317
FederatedSelect	0.8257	0.8759	0.9064	0.9101
NeuLite	0.7965	0.8665	0.9075	0.9100

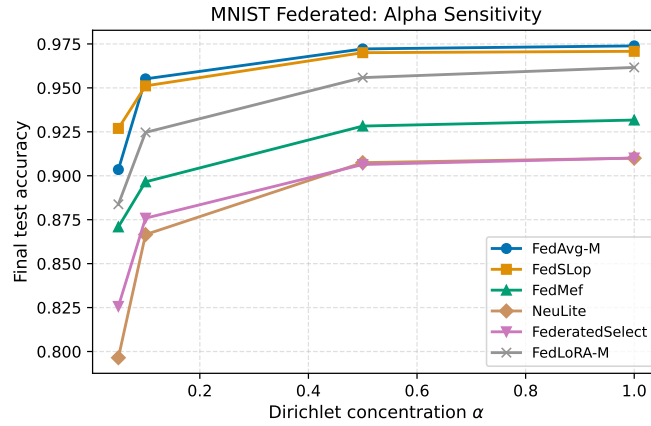


Figure 2: Sensitivity to the Dirichlet concentration parameter α . Final test accuracy after 100 communication rounds.

5.3 Robustness to Statistical Heterogeneity

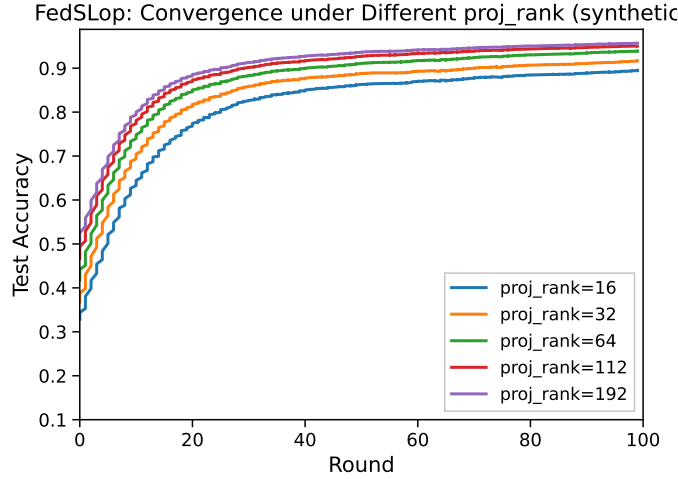
To study sensitivity to the non-IID level, we vary the Dirichlet concentration parameter. The results are summarized in Table 3 and Figure 2. As α decreases, all methods deteriorate, but the extent of deterioration differs substantially.

The most notable case is the extreme non-IID regime $\alpha = 0.05$: FEDSLOP reaches 0.9269, while FEDAVGWITHMOMENTUM drops to 0.9035. Thus the proposed method is more robust than the strongest full-parameter baseline under severe heterogeneity. For milder heterogeneity levels ($\alpha \geq 0.1$), FEDAVGWITHMOMENTUM retains a small advantage, but the difference remains modest.

These results suggest that the projected update structure of FEDSLOP acts as an effective regularizer in highly heterogeneous regimes. The sparse and blockwise baselines degrade much more sharply, while the LoRA baseline lies in between: it is more stable than the sparse baselines but still clearly below FEDSLOP in the hardest setting.

Table 4: Rank ablation for FEDSLOP on MNIST with $\alpha=0.1$ after 100 communication rounds.

Projection rank r	Final accuracy (mean $\pm$ std)
16	0.8907 ± 0.0113
32	0.9148 ± 0.0055
64	0.9392 ± 0.0035
112	0.9511 ± 0.0028
192	0.9561 ± 0.0034

Figure 3: Convergence of FEDSLOP for different projection ranks r .

5.4 Projection-Rank Ablation

The projection rank r is the central design parameter of FEDSLOP. Table 4 and Figure 3 report the effect of varying $r \in \{16, 32, 64, 112, 192\}$ under the same MNIST setup.

The trend is monotone and intuitive: increasing r substantially improves performance from $r = 16$ to $r = 112$, while the gain from $r = 112$ to $r = 192$ is relatively small. Hence $r = 112$ already captures most of the useful optimization subspace in this MNIST task.

5.5 Scaling with the Number of Clients

We next fix the projection rank at $r = 112$ and vary the number of clients. Table 5 and Figure 4 summarize the results.

Increasing the number of clients from 10 to 50 noticeably improves both convergence speed and final accuracy. However, the gain from 50 to 100 clients is marginal, indicating a diminishing-returns regime in which additional client participation mainly increases communication cost without bringing comparable accuracy improvements.

Table 5: Client-number ablation for FEDSLOP on MNIST with $\alpha=0.1$ after 100 communication rounds.

Number of clients p	Total uplink elements	Final accuracy (mean $\pm$ std)
10	203,540	0.9199 ± 0.0094
20	407,080	0.9341 ± 0.0089
50	1,017,700	0.9472 ± 0.0026
100	2,035,400	0.9479 ± 0.0010

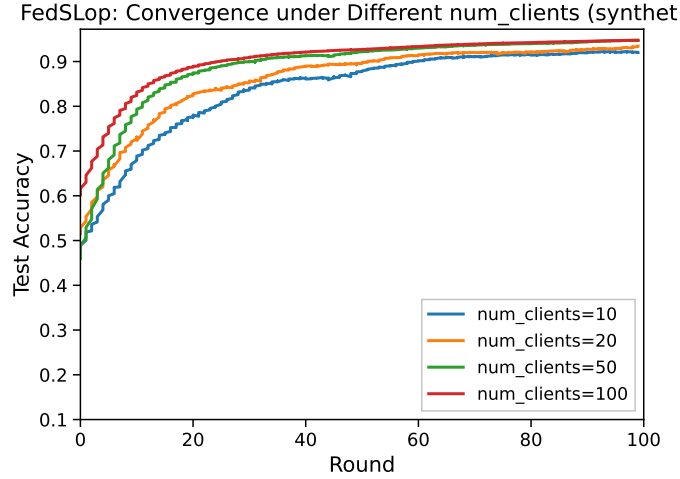


Figure 4: Convergence of FEDSLOP for different numbers of clients.

5.6 Summary

Across all experiments, FEDSLOP achieves a strong balance between accuracy and communication efficiency. On non-IID MNIST it remains very close to the full-parameter FEDAVGWITHMOMENTUM baseline, substantially outperforms sparse and blockwise update methods, and shows markedly better robustness in the strongly heterogeneous regime. The rank and client-number ablations further indicate that moderate projection rank and moderate client population already capture most of the attainable benefit in this task.

6 Conclusion

We have proposed FedSLop, a FedAvg-style federated learning algorithm that integrates random low-rank subspace projections with momentum in order to reduce client-side memory usage while retaining the simplicity and robustness of the standard FL workflow. By sampling a fresh Stiefel subspace at each communication round and constraining all local stochastic gradients and momentum updates to this subspace, FedSLop reduces

the effective dimensionality of the optimizer state and the communicated directions without modifying the underlying model architecture.

On the theoretical side, we developed a detailed nonconvex convergence analysis for FedSLoP under full client participation and random Stiefel projectors. Under standard smoothness and bounded-variance assumptions, we derived explicit bounds on the averaged squared gradient norm that exhibit an $\mathcal{O}(1/T)$ optimization term and an $\mathcal{O}(T^{-1/2})$ variance term, with constants that depend explicitly on the subspace ratio $\underline{\delta} = r/d$, the number of local steps τ , the momentum parameter μ , and the number of clients N . The resulting variance floor contains a linear speedup factor $1/N$ in the local variance term, and a harmonic step-size strategy yields a provably near-optimal balance between optimization, variance, and drift terms.

On the empirical side, our experiments on federated MNIST classification with Dirichlet non-IID partitions showed that FedSLoP closely tracks the convergence of a strong FedAvgWithMomentum baseline while substantially outperforming sparse and low-rank model baselines such as FedMef, NeuLite, FederatedSelect, and FedLoRAWithMomentum. Projection-rank and client-number ablations revealed clear diminishing-returns regimes and confirmed that moderate ranks and client counts already suffice to capture the dominant gradient subspace and achieve stable, accurate training under severe heterogeneity.

Taken together, these results demonstrate that constraining the optimizer state to random low-dimensional subspaces is a principled and effective strategy for communication- and memory-efficient federated learning. Several directions remain for future work. Extending the convergence analysis to partial participation and asynchronous settings, incorporating adaptive or data-dependent subspace selection, and applying FedSLoP to larger-scale models and tasks (e.g., language modeling and vision transformers) are particularly promising. Another avenue is to combine the subspace-based design with advanced compression and personalization techniques, further closing the gap between theoretical guarantees and the demanding system constraints of real-world FL deployments.

References

- [1] Dan Alistarh, Demjan Grubic, Jerry Li, Ryota Tomioka, and Milan Vojnovic. Qsgd: Communication-efficient SGD via gradient quantization and encoding. *arXiv preprint arXiv:1610.02132v4*, 2016.
- [2] Debraj Basu, Deepesh Data, Can Karakus, and Suhas Diggavi. Qsparse-local-SGD: Distributed SGD with quantization, sparsification, and local computations. *IEEE Journal on Selected Areas in Information Theory*, 2019.
- [3] Aleksandr Beznosikov, Samuel Horváth, Peter Richtárik, and Mher Safaryan. On biased compression for distributed learning. *Journal of Machine Learning Research*, 2020.
- [4] Ziheng Cheng, Xinmeng Huang, Pengfei Wu, and Kun Yuan. Momentum benefits non-iid federated learning simply and provably. In *Proceedings of the International Conference on Learning Representations*, 2023.

- [5] Laurent Condat, Artavazd Maranjyan, and Peter Richtárik. Locodl: Communication-efficient distributed learning with local training and compression. In *International Conference on Learning Representations*, 2024.
- [6] DeepSeek-AI, Aixin Liu, Bei Feng, Bin Wang, Bingxuan Wang, Bo Liu, Chenggang Zhao, Chengqi Deng, Chong Ruan, Damai Dai, Daya Guo, Dejian Yang, Deli Chen, Dongjie Ji, Erhang Li, Fangyun Lin, Fuli Luo, Guangbo Hao, Guanting Chen, Guowei Li, H. Zhang, Hanwei Xu, Hao Yang, Haowei Zhang, Honghui Ding, Huajian Xin, Huazuo Gao, Hui Li, Hui Qu, J. L. Cai, Jian Liang, Jianzhong Guo, Jiaqi Ni, Jiashi Li, Jin Chen, Jingyang Yuan, Junjie Qiu, Junxiao Song, Kai Dong, Kaige Gao, Kang Guan, Lean Wang, Lecong Zhang, Lei Xu, Leyi Xia, Liang Zhao, Liyue Zhang, Meng Li, Miaojun Wang, Mingchuan Zhang, Minghua Zhang, Minghui Tang, Mingming Li, Ning Tian, Panpan Huang, Peiyi Wang, Peng Zhang, Qihao Zhu, Qinyu Chen, Qiushi Du, R. J. Chen, R. L. Jin, Ruiqi Ge, Ruizhe Pan, Runxin Xu, Ruyi Chen, S. S. Li, Shanghao Lu, Shangyan Zhou, Shanhuang Chen, Shaoqing Wu, Shengfeng Ye, Shirong Ma, Shiyu Wang, Shuang Zhou, Shuiping Yu, Shunfeng Zhou, Size Zheng, T. Wang, Tian Pei, Tian Yuan, Tianyu Sun, W. L. Xiao, Wangding Zeng, Wei An, Wen Liu, Wenfeng Liang, Wenjun Gao, Wentao Zhang, X. Q. Li, Xiangyue Jin, Xianzu Wang, Xiao Bi, Xiaodong Liu, Xiaohan Wang, Xiaojin Shen, Xiaokang Chen, Xiaosha Chen, Xiaotao Nie, Xiaowen Sun, Xiaoxiang Wang, Xin Liu, Xin Xie, Xingkai Yu, Xinnan Song, Xinyi Zhou, Xinyu Yang, Xuan Lu, Xuecheng Su, Y. Wu, Y. K. Li, Y. X. Wei, Y. X. Zhu, Yanhong Xu, Yanping Huang, Yao Li, Yao Zhao, Yaofeng Sun, Yaohui Li, Yaohui Wang, Yi Zheng, Yichao Zhang, Yiliang Xiong, Yilong Zhao, Ying He, Ying Tang, Yishi Piao, Yixin Dong, Yixuan Tan, Yiyuan Liu, Yongji Wang, Yongqiang Guo, Yuchen Zhu, Yudian Wang, Yuheng Zou, Yukun Zha, Yunxian Ma, Yuting Yan, Yuxiang You, Yuxuan Liu, Z. Z. Ren, Zehui Ren, Zhangli Sha, Zhe Fu, Zhen Huang, Zhen Zhang, Zhenda Xie, Zhewen Hao, Zhihong Shao, Zhiniu Wen, Zhipeng Xu, Zhongyu Zhang, Zhuoshu Li, Zihan Wang, Zihui Gu, Zilin Li, and Ziwei Xie. DeepSeek-V2: A strong, economical, and efficient mixture-of-experts language model. *arXiv preprint arXiv:2405.04434v5*, 2024.
- [7] Haoze He, Jing Wang, and Anna Choromanska. Adjacent leader decentralized stochastic gradient descent. In *Proceedings of the European Conference on Artificial Intelligence*, 2024.
- [8] Yutong He, Xinmeng Huang, Yiming Chen, Wotao Yin, and Kun Yuan. Lower bounds and accelerated algorithms in distributed stochastic optimization with communication compression. *arXiv preprint arXiv:2305.07612*, 2023.
- [9] Yutong He, Xinmeng Huang, and Kun Yuan. Unbiased compression saves communication in distributed optimization: When and how much? In *Advances in Neural Information Processing Systems*, 2023.
- [10] Yutong He, Pengrui Li, Yipeng Hu, Chuyan Chen, and Kun Yuan. Subspace optimization for large language models with convergence guarantees. *arXiv preprint arXiv:2410.11289*, 2024.
- [11] Samuel Horváth, Dmitry Kovalev, Konstantin Mishchenko, Sebastian Stich, and Peter Richtárik. Stochastic distributed learning with gradient quantization and variance reduction. *arXiv preprint arXiv:1904.05115*, 2019.
- [12] Like Jian and Dong Liu. Widening the network mitigates the impact of data heterogeneity on FedAvg. In *Proceedings of the International Conference on Machine Learning*, 2025.
- [13] Sai Praneeth Karimireddy, Satyen Kale, Mehryar Mohri, Sashank J. Reddi, Sebastian U. Stich, and Ananda Theertha Suresh. SCAFFOLD: Stochastic controlled averaging for federated learning. *arXiv preprint arXiv:1910.06378v4*, 2019.
- [14] Sai Praneeth Karimireddy, Quentin Rebjock, Sebastian U. Stich, and Martin Jaggi. Error feedback fixes signsgd and other gradient compression schemes. In *Proceedings of the Inter-*

national Conference on Machine Learning, 2019.

- [15] Xiang Li, Kaixuan Huang, Wenhao Yang, Shusen Wang, and Zhihua Zhang. On the convergence of FedAvg on non-IID data. In *Proceedings of the International Conference on Learning Representations*, 2019.
- [16] Xiangru Lian, Wei Zhang, Ce Zhang, and Ji Liu. Asynchronous decentralized parallel stochastic gradient descent. In *Proceedings of the International Conference on Machine Learning*, 2017.
- [17] Junkang Liu, Fanhua Shang, Junchao Zhou, Hongying Liu, Yuanyuan Liu, and Jin Liu. Fedmuon: Accelerating federated learning with matrix orthogonalization. *arXiv preprint arXiv:2510.27403v1*, 2025.
- [18] Klaus Neymeyr. A geometric convergence theory for the preconditioned steepest descent iteration. *SIAM Journal on Numerical Analysis*, 2011.
- [19] Yaqian Qi, Yuan Feng, Xiangxiang Wang, Hanzhe Li, and Jingxiao Tian. Leveraging federated learning and edge computing for recommendation systems within cloud computing networks. *arXiv preprint arXiv:2403.03165v2*, 2024.
- [20] Xun Qian, Peter Richtárik, and Tong Zhang. Error compensated distributed sgd can be accelerated. In *Advances in Neural Information Processing Systems*, 2020.
- [21] Zhaonan Qu, Kaixiang Lin, Zhaojian Li, Jiayu Zhou, and Zhengyuan Zhou. A unified linear speedup analysis of federated averaging and nesterov FedAvg. *Journal of Artificial Intelligence Research*, 2020.
- [22] Hamza Reguieg, Mohammed El Hanjri, Mohamed El Kamili, and Abdellatif Kobbane. A comparative evaluation of fedavg and per-fedavg algorithms for dirichlet distributed heterogeneous data. In *International Conference on Wireless Networks and Mobile Communications*, 2023.
- [23] Sebastian U. Stich, Jean-Baptiste Cordonnier, and Martin Jaggi. Sparsified sgd with memory. In *Advances in Neural Information Processing Systems*, 2018.
- [24] Sebastian U. Stich and Sai Praneeth Karimireddy. The error-feedback framework: Better rates for SGD with delayed gradients and compressed communication. *arXiv preprint arXiv:1909.05350*, 2019.
- [25] Lili Su, Jiaming Xu, and Pengkun Yang. A non-parametric view of FedAvg and FedProx: Beyond stationary points. *Journal of Machine Learning Research*, 2021.
- [26] Yuki Takezawa and Sebastian U. Stich. Scalable decentralized learning with teleportation. In *Proceedings of the International Conference on Learning Representations*, 2025.
- [27] Hanlin Tang, Xiangru Lian, Ming Yan, Ce Zhang, and Ji Liu. D²: Decentralized training over decentralized data. In *Proceedings of the International Conference on Machine Learning*, 2018.
- [28] Hanlin Tang, Xiangru Lian, Chen Yu, Tong Zhang, and Ji Liu. DoubleSqueeze: Parallel stochastic gradient descent with double-pass error-compensated compression. In *Proceedings of the International Conference on Machine Learning*, 2019.
- [29] Quoc Tran-Dinh, Nhan H. Pham, Dzung T. Phan, and Lam M. Nguyen. FedDR – randomized douglas-rachford splitting algorithms for nonconvex federated composite optimization. In *Advances in Neural Information Processing Systems*, 2021.
- [30] Anastasiia Usmanova, François Portet, Philippe Lalanda, and German Vega. A distillation-based approach integrating continual learning and federated learning for pervasive services. *arXiv preprint arXiv:2109.04197v1*, 2021.
- [31] Thijs Vogels, Sai Praneeth Karimireddy, and Martin Jaggi. Powersgd: Practical low-rank gradient compression for distributed optimization. In *Advances in Neural Information Processing Systems*, 2019.

- [32] Yingchao Wang and Wenqi Niu. Federated progressive self-distillation with logits calibration for personalized iiot edge intelligence. *arXiv preprint arXiv:2412.00410v1*, 2024.
- [33] Ming Xiang, Stratis Ioannidis, Edmund Yeh, Carlee Joe-Wong, and Lili Su. Towards bias correction of FedAvg over nonuniform and time-varying communications. In *Proceedings of the IEEE Conference on Decision and Control*, 2023.
- [34] Ran Xin, Usman A. Khan, and Soummya Kar. An improved convergence analysis for decentralized online stochastic non-convex optimization. *IEEE Transactions on Signal Processing*, 2020.
- [35] Haibo Yang, Minghong Fang, and Jia Liu. Achieving linear speedup with partial worker participation in non-IID federated learning. In *Proceedings of the International Conference on Learning Representations*, 2021.
- [36] Fuxun Yu, Weishan Zhang, Zhuwei Qin, Zirui Xu, Di Wang, Chenchen Liu, Zhi Tian, and Xiang Chen. Heterogeneous federated learning. *arXiv preprint arXiv:2008.06767v2*, 2020.
- [37] Honglin Yuan, Manzil Zaheer, and Sashank Reddi. Federated composite optimization. *arXiv preprint arXiv:2011.08474v3*, 2020.
- [38] Kun Yuan, Sulaiman A. Alghunaim, and Xinmeng Huang. Removing data heterogeneity influence enhances network topology dependence of decentralized SGD. *Journal of Machine Learning Research*, 2021.
- [39] Hao Zhang, Tingting Wu, Siyao Cheng, and Jie Liu. CC-FedAvg: Computationally customized federated averaging. *IEEE Internet of Things Journal*, 2022.
- [40] Jiaojiao Zhang, Jiang Hu, and Mikael Johansson. Non-convex composite federated learning with heterogeneous data. *arXiv preprint arXiv:2502.03958v1*, 2025.
- [41] Jiaojiao Zhang, Yuqi Xu, and Kun Yuan. An efficient subspace algorithm for federated learning on heterogeneous data. *arXiv preprint arXiv:2509.05213*, 2025.
- [42] Yuan Zhou, Jiachen Zhong, Xinli Shi, Guanghui Wen, and Xinghuo Yu. FedCanon: Non-convex composite federated learning with efficient proximal operation on heterogeneous data. *IEEE Transactions on Signal Processing*, 2025.